

CS 660: Data Mining for Decision Making

Course Code: CS 660

Course Name: Data Mining for Decision Making

L-T-P-C: 4-0-0-4

Prerequisites: IC 210: Probability, Statistics and Random Processes; IC 250: Data Structure and Algorithms

Intended for: UG/PG

Distribution: Discipline elective for CSE; CS elective for EE and ME

Approval: 8th Senate

Course Contents

The course is divided into weekly modules, where a new topic is covered in each week. The details of the topics covered in each week are provided below:

- Week 1: **Introduction to Data mining:** What is Data Mining? What is the Data Mining Process? Basic Data Mining Tasks, Problem Identification, Data Mining Metrics, Data Cleaning (pre-processing, feature selection, data reduction, feature encoding, noise and missing values, etc.), Key Issues, Opportunities for Data Mining. [3 Lectures]
- Week 2: **Nave Bayes classifier:** Two-class classifiers, Training and Test sets, Maximum-Likelihood estimation, Bayesian estimation, Classification of Test sets. [3 Lectures]
- Week 3: **Decision Trees:** Classification and Regression Trees, Building and Selecting Decision Trees (concept of Information Gain), Obtaining Production Rules from Decision Trees, Handling missing values in Decision Trees. [3 Lectures]
- Week 4: **Neural Networks:** Introduction to Artificial Neural Networks, Single-layer Networks, Multi-layer Networks, Backward Propagation Algorithm, Annealing the learning rate (Step decay, Adagrad, and RMSprop), Over-fitting and choice of Epochs. [3 Lectures]
- Week 5: **Instance-Based Learning:** Instances, Activations, Recency, Frequency, Retrieval from Memory, Blending of instances. [3 Lectures]
- Week 6: **Clustering:** Introduction to Cluster Analysis, Clustering Algorithms, Hierarchical Methods (Nearest neighbor, Farthest neighbor, Group average), Similarity Measures. [3 Lectures]
- Week 7: **Logistic Regression:** Introduction to Logistic Regression, Logistic function, odds ratio, logit, Logistic Regression with more than two classes. [3 Lectures]
- Week 8: **Multiple-Linear Regression:** Introduction to Multiple-Linear Regression, Assumption made in a linear regression model, regression process, dropping irrelevant variables. [3 Lectures]
- Week 9: **Principal Components Analysis:** Introduction to principal components analysis, dimensionality reduction, principal components and orthogonal least squares. [3 Lectures]
- Week 10: **Discriminant Analysis:** Introduction to discriminant analysis, applications to two-classes, extension to more than 2- classes, canonical variate loadings, extension to unequal covariance structures. [3 Lectures]
- Week 11: **Association Rules:** Introduction to association rules, support, confidence, Apriori Algorithm. [3 Lectures]



- Week 13: **Implementation Issues:** Metrics for Model selection - MDL, BIC, AIC, Ethics, Legality, and Privacy; Staffing and Implementation [3 Lectures]
- Week 14: **The Future of Data Mining, Unstructured Data Mining, and conclusions:** [3 Lectures]
- **If time permits:** Topics in graph mining: Definition of Graphs, Subgraphs, Frequent Subgraphs and subgraphs, Detection Algorithms: Apriori-Based Approach, Pattern Growth Approach (gSpan), Graph Classification, and Graph Compression.

Textbooks:

1. Hand David, Mannila Heikki, and Smyth Padhraic, **Principles of Data Mining**, MIT, 2004.,

References:

1. Han, J., Kamber, M. & Pei, J., **Data mining concepts and techniques**, 3rd Edition Morgan Kaufmann Publishers, 2012.
2. Berry and Linoff, **Mastering Data Mining**, Wiley, 2000.
3. Delmater and Hancock, **Data Mining Explained**, Digital Press, 2001.
4. T. Mitchell, **Machine Learning**, McGraw-Hill, 1997.
5. M. H. Dunham, **Data Mining: Introductory and Advanced Topics**, Pearson Education, 2001.
6. Samatova, N. F., Hendrix, W., Jenkins, J., Padmanabhan, K., & Chakraborty, A. (Eds.), **Practical Graph Mining with R**, CRC Press, 2013.
7. Wang, H., **Managing and mining graph data** (Vol. 40), C. C. Aggarwal (Ed.), Springer, 2010.